MILITARY STANDARD

MIL-STD-1320-150
(Navy)

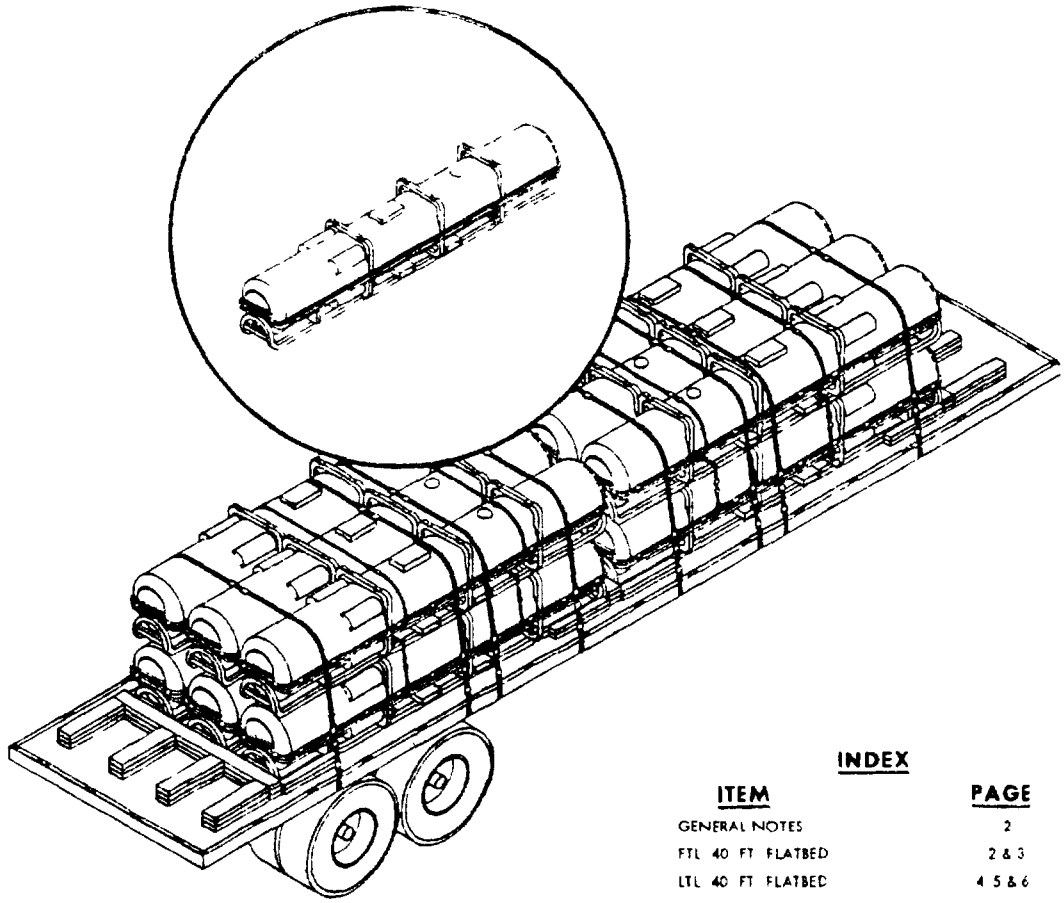
25 MAY 1973

SUPERSEDING
WR-51/150
28 JANUARY 1972

TRUCKLOADING
GUIDED WEAPON MK 5 & MODS (WALLEYE)
IN CONTAINER CNU-154 A/E OR CNU-154 B/E

CONTAINER DATA

	CNU-154 A/E	CNU-154 B/E
WEIGHT EMPTY	650 LBS	610 LBS
WEIGHT LOADED	2950 LBS	2910 LBS
DIMENSIONS	182 3/4 L x 28 3/4 W x 37 2/32 H	172 L x 28 3/4 W x 32 21/32 H
STACKING HEIGHT	32 11/32	32 11/32
CUBE	99 5 CU FT	93 5 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES A	EXPLOSIVES A



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 FT FLATBED	2 & 3
LTL 40 FT FLATBED	4 5 & 6
UNITIZING	7
LOAD STRAPPING	7

NOTES

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

2 FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M S Gray NWHL 4/12/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J C Hill 4/20/73
SIGNATURE (AIR SYSCOM BY DIRECTION DATE

ORIGINATOR Charles M Wade 4/17/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1 THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE GUIDED WEAPON MK 5 AND MODS (WALLEYE) IN CONTAINER CNU-154 A/E OR CNU-154 B/E

2 THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER) DO NOT USE TRAILERS WITH METAL FLOORS

3 WHEN CONTAINERS ARE TWO HIGH THEY SHALL BE UNITIZED AS SHOWN ON PAGE 7

WARNING

THE BOTTOM LAYER OF EACH STACK MUST BE OF LIKE CONTAINERS

4 CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS

5 STEEL STRAPS SHALL BE 2 x .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENTS SHOWN ON PAGE 7

6 WHEN USING CHAINS FOR TIE-DOWNS, THE LADING SHALL BE PROTECTED FROM CHAIN DAMAGE WHERE THE CHAIN GOES OVER THE TOP OF THE LOAD BY INSERTING A DOUBLED 2 x 6 BETWEEN THE CHAIN AND THE LOAD

7 CHAINS AND LOADBINDERS SHALL MEET THE FOLLOWING CONDITIONS

A CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY

B CHAINS FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS

C LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN

8 PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626 THE CHAINS, FITTINGS, AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO 22 OF DD FORM 626 ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER

9 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

10 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO DUNNAGE IN AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER

11 APPLICABLE MATERIAL SPECIFICATIONS

DUNNAGE LUMBER, MM-L-751

NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT

STRAPPING, QQ-S-781, TYPE I CLASS A

12 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY)

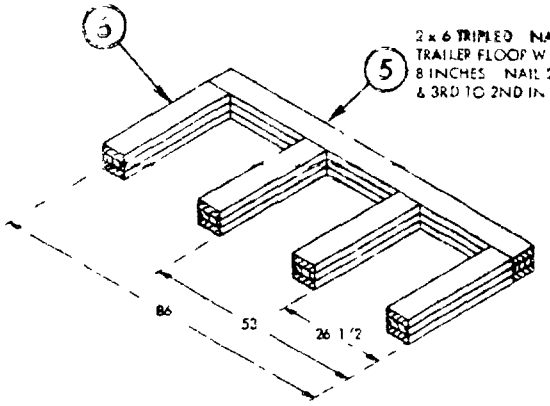
FTL 40 FT TRAILER (FLATBED)

1 CONTAINERS SHOULD BE STACKED TWO HIGH AND STRAPPED TOGETHER BEFORE LOADING ON TRAILER, FOR UNITIZING DETAILS, SEE PAGE 7

2 STRAP TOP LAYER OF EACH STACK OF CONTAINERS TOGETHER WITH TWO 1 1/4 STRAPS (PIECE NO 7) IN POSITION SHOWN AFTER TENSIONING STRAPS SEAL WITH TWO 1 1/4" SEALS (PIECE NO 2) DOUBLE CRIMP SEALS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

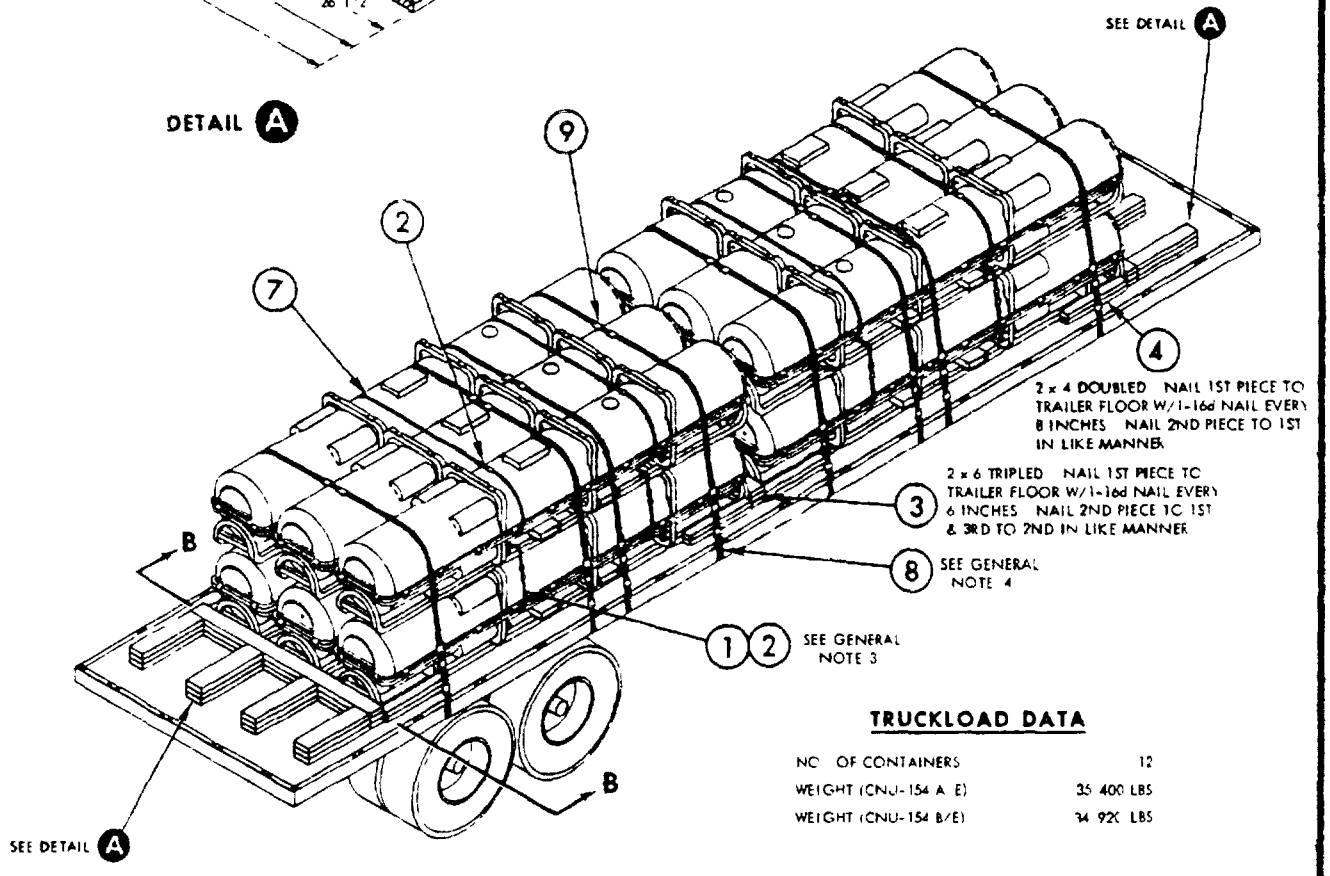
2 x 6 TRIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 6 INCHES NAIL 2ND PIECE TO 1ST & 3RD TO 2ND IN LIKE MANNER



NOTE

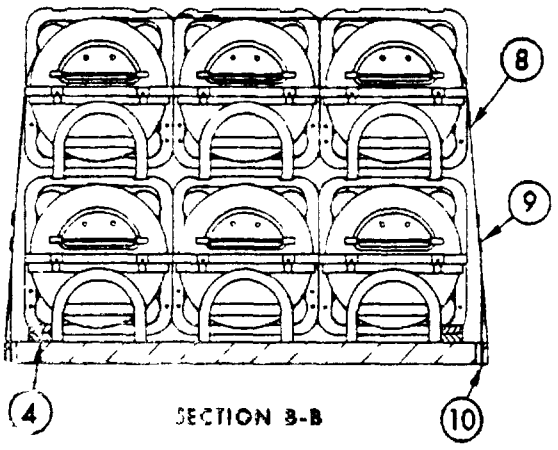
THIS FTL ILLUSTRATES THE USE OF 2 x .050 STRAPS AS TIEDOWN. CHAINS ARE AN AUTHORIZED ALTERNATE SEE GENERAL NOTE 4

DETAIL A



TRUCKLOAD DATA

NO OF CONTAINERS	12
WEIGHT (CNU-154 A/E)	35 400 LBS
WEIGHT (CNU-154 B/E)	34 920 LBS



SECTION B-B

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	STAKE POCKET PAD	STEEL STRAP 2 x .050 x 16	16	(MAY USE COMMERCIAL STAKE POCKET PAD IF AVAILABLE)		
9	SEAL	2 x 22 GA MIN	64	-		-
8	STRAP	2 x .050	8	-	-	-
7	CROSS STRAP	1 1/4 x .035	4	-	-	-
6	BACKUP CLEAT	2 x 6 x 30	24	SEE NOTE DETAIL A		
5	END CROSSMEMBER	2 x 6 x 86	6	SEE NOTE DETAIL A		
4	SLEEPER	2 x 4 x 30	16	SEE FIELD NOTE		
3	CENTER CROSSMEMBER	2 x 6 x 86	3	SEE FIELD NOTE		
2	SEAL	1 1/4 x 22 GA MIN	32	-	-	-
1	STRAP	1 1/4 x .035	12	-	-	-

LIST OF MATERIALS & NAILING DATA

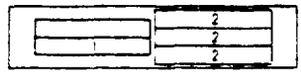
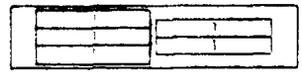
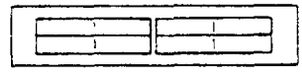
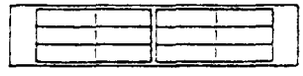
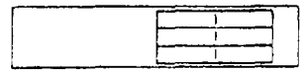
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

LTL 40 FT TRAILER (FLATBED)

1 WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED FROM TABLE I AND POSITION CONTAINERS ON THE TRAILER ACCORDINGLY

TABLE I

LOAD PATTERN*

NO OF CONTAINERS	→ FWD	NO OF CONTAINERS	→ FWD	NO OF CONTAINERS	→ FWD
11		8		5	
10		7		4	
9		6		3	

*LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER NUMBER INDICATES THE NUMBER OF CONTAINERS IN A LAYER

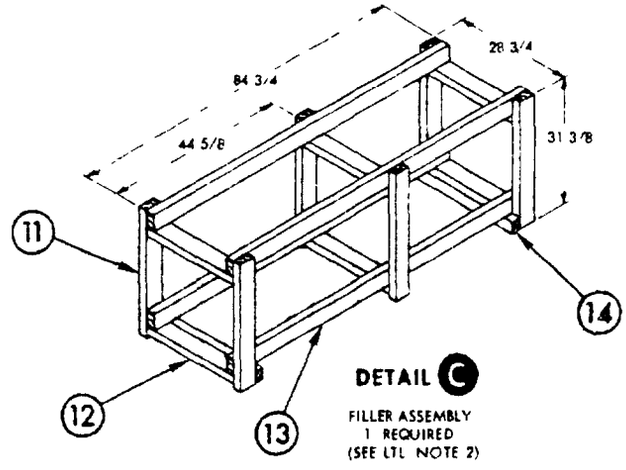
2 WHEN THE CENTER OF THE TOP LAYER OF A STACK DOES NOT HAVE A CONTAINER (SEE LOAD PLAN FOR 11 CONTAINERS), THE FILLER ASSEMBLY DETAIL C SHALL BE USED THE ASSEMBLY SHALL BE STRAPPED TO THE LOWER CONTAINER SIMILAR TO THE PROCEDURE SHOWN ON PAGE 7 PIECES 11 OF DETAIL C SHOULD BE OPPOSITE CONTAINER ROLL RINGS

3 ALL STACKS OF TWO HIGH CONTAINERS SHALL BE SECURED TO THE TRAILER WITH FOUR STRAPS (OR CHAINS) STACKS OF CONTAINERS ONE HIGH SHALL BE SECURED WITH THREE STRAPS (OR CHAINS)

4 THE LTL SHOWN ON PAGE 5 IS FOR 11 CONTAINERS AND SHOWS THE CORRECT BLOCKING BRACING AND USE OF TIE-DOWN STRAPS FOR A TRUCKLOAD OF THIS QUANTITY

5 THE LTL SHOWN ON PAGE 6 IS FOR EIGHT CONTAINERS AND SHOWS THE CORRECT BLOCKING, BRACING AND USE OF TIE-DOWN CHAINS FOR A TRUCKLOAD OF THIS QUANTITY

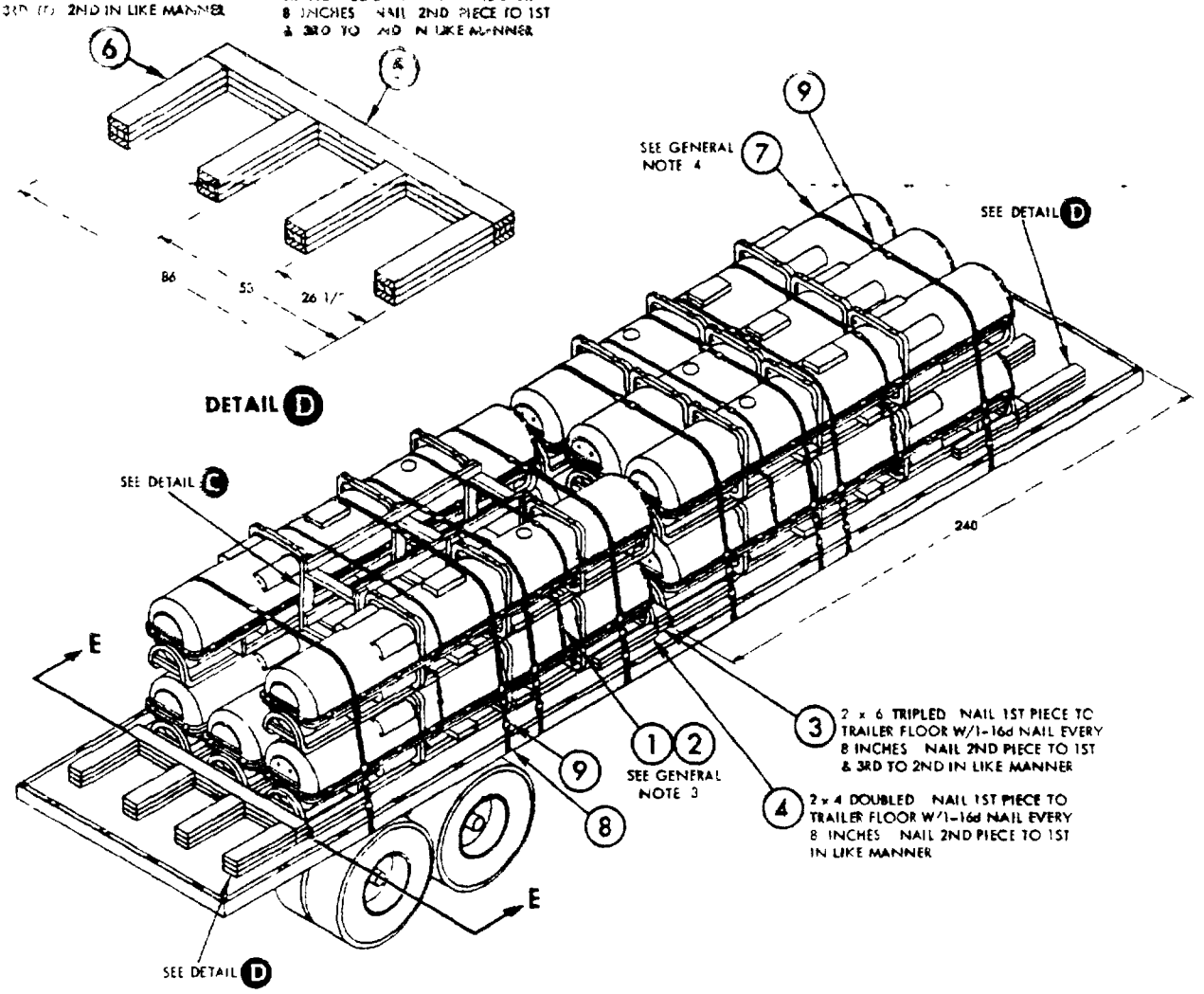
6 THE LTLS SHOWN ON PAGES 5 AND 6 DO NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY THEIR PRESENTATION IS INTENDED TO EXPLAIN THE CORRECT METHODS OF LOADING LTL SHIPMENTS



DETAIL C

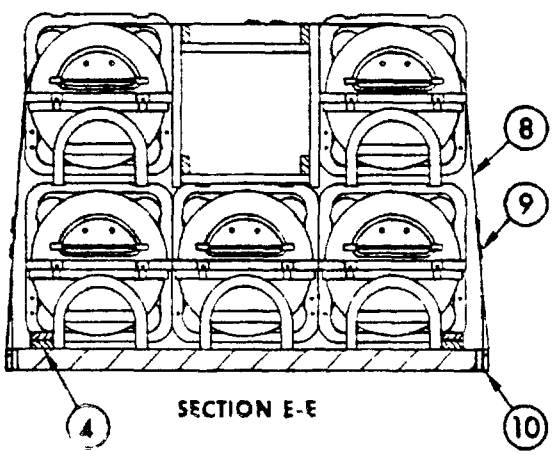
FILLER ASSEMBLY
1 REQUIRED
(SEE LTL NOTE 2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.



NOTE

THIS LTL ILLUSTRATES THE USE OF 2 x .050 STRAPS AS TIEDOWNS. CHAINS ARE AN AUTHORIZED ALTERNATE. SEE GENERAL NOTE 4.



SECTION E-E

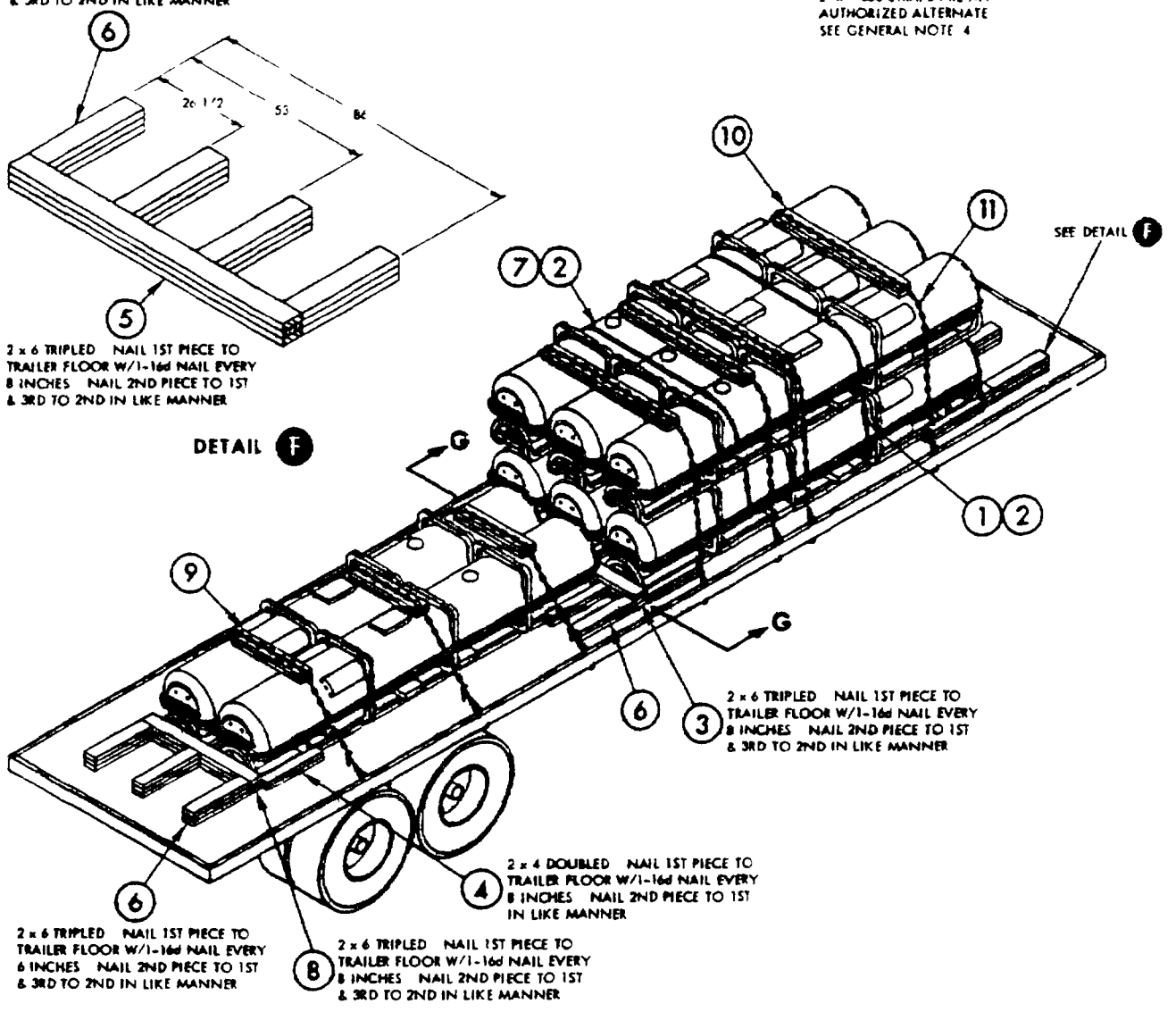
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
14	STOP	2 x 3 x 28 3/4	2	12 / 1	4 / 2 JOINT	12c / 12a
13	LONGITUDINAL	2 x 4 x 84 3/4	4	SEE 1		-
12	STRUT	2 x 4 x TO SUIT	6	SEE 11	-	-
11	VERTICAL	2 x 4 x 31 3/8	6	12 & 13	3/JOINT	12d
10	STAKE POCKET PAD	STEEL STRAP 2 x .050 x 18	16	(MAY USE COMMERCIAL STAKE POCKET PAD IF AVAILABLE)		
9	SEAL	2 x 22 GA MIN	64	-	-	-
8	STRAP	2 x .050	8	-	-	-
7	CROSS STRAP	1 1/4 x .035	4	-	-	-
6	BACKUP CLEAT	2 x 6 x 30	24	SEE NOTE DETAIL A		-
5	END CROSSMEMBER	2 x 6 x 86	6	SEE NOTE DETAIL A		-
4	SLEEPER	2 x 4 x 30	16	SEE FIELD NOTE		-
3	CENTER CROSSMEMBER	2 x 6 x 86	3	SEE FIELD NOTE		-
2	SEAL	1 1/4 x 22 GA MIN	32	-	-	-
1	STRAP	1 1/4 x .035	12	-	-	-

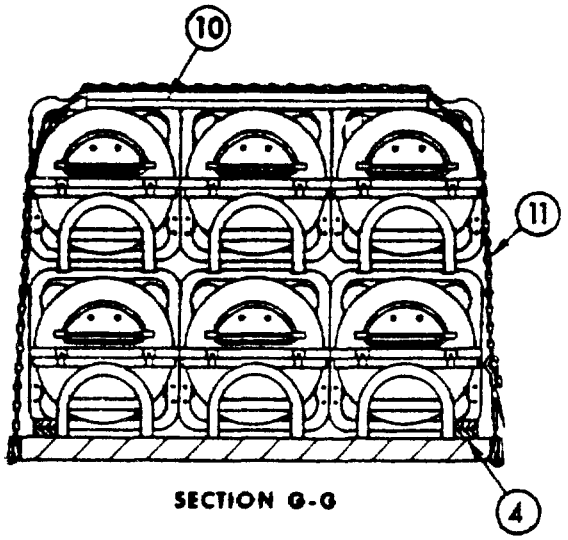
LIST OF MATERIALS & NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

NOTE

THIS LTL ILLUSTRATES THE USE OF CHAINS AND LOAD BINDERS 2 x .050 STRAPS ARE AN AUTHORIZED ALTERNATE SEE GENERAL NOTE 4





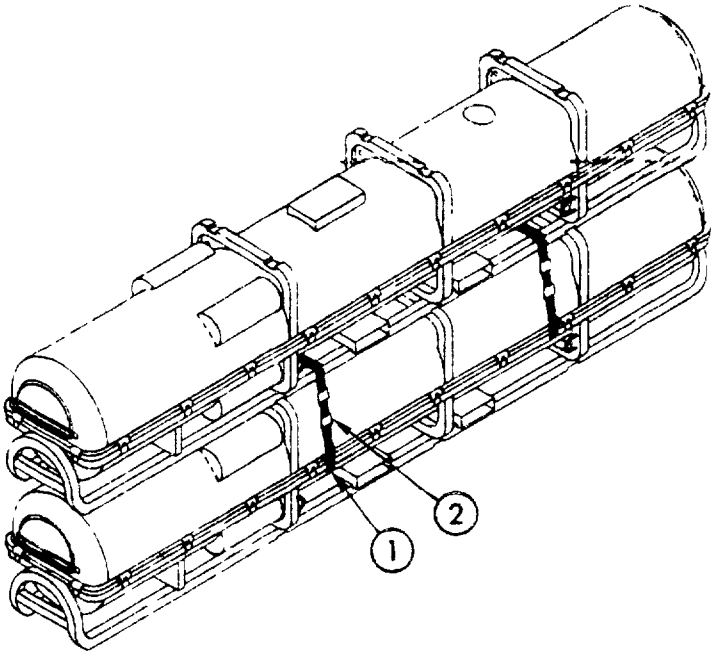
SECTION G-G

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
11	CHAIN & LOAD BINDER	5/8	7	-	-	-
10	PROTECTION BOARD	2 x 6 x 65	8	DOUBLED & LAMINATED	2/FOOT	12d
9	PROTECTION BOARD	2 x 6 x 36	6	DOUBLED & LAMINATED	2/FOOT	12d
8	REAR CROSSMEMBER	2 x 6 x 58	3	SEE FIELD NOTE		
7	CROSS STRAP	1 1/4 x .035	2	-	-	-
6	BACKUP CLEAT	2 x 6 x 30	27	SEE FIELD NOTE		
5	FORWARD CROSSMEMBER	2 x 6 x 86	3	SEE DETAIL F		
4	SLEEPER	2 x 4 x 30	16	SEE FIELD NOTE		
3	CENTER CROSSMEMBER	2 x 6 x 86	3	SEE FIELD NOTE		
2	SEAL	1 1/4 x 22 GA MIN	16	-	-	-
1	UNITIZING STRAP	1 1/4 x .035	6	-	-	-

LIST OF MATERIALS & NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.

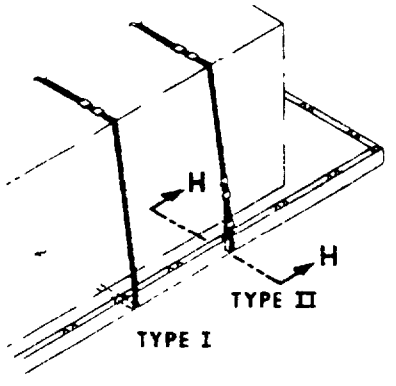
UNITIZING

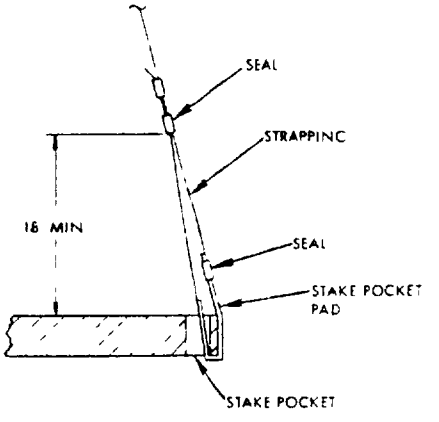


CONTAINERS STACKING AND BANDING PROCEDURE

1. STACK TWO CONTAINERS USING A FORK LIFT TRUCK OR SUITABLE HOISTING DEVICE EQUIPPED WITH A SUITABLE FOUR LEG BRIDLE
2. BE CERTAIN THAT THE NESTING FEATURES AND BEARING SURFACES ARE PROPERLY POSITIONED TO PREVENT SHIFTING IN LATERAL AND LONGITUDINAL DIRECTIONS
3. SECURE BOTTOM CONTAINER TO TOP CONTAINER WITH TWO 1 1/4 x .035 STEEL STRAPS (PIECE NO 1), POSITIONED AS INDICATED ABOVE
4. AFTER TENSIONING STRAPS, SEAL WITH TWO 1 1/4 SEALS (PIECE NO 2) DOUBLE CRIMP SEALS

LOAD STRAPPING





SECTION H-H

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAP JOINT SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PERCENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD.

TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH THE DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION H-H. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.